

# Mika Be

## APPLIED AI | SOFTWARE ENGINEERING

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## EDUCATION

**Rutgers University, New Brunswick** | B.S. Electrical and Computer Engineering, CS minor Expected May 2028

**GPA** 3.838/4.000 | Dean's List, all four completed semesters

**Honors** 2nd Place Best Full Paper, IEEE ISEC 2026 | IEEE-HKN honor society | Bergen County Academies CS Award (1 of 3)

**Coursework** Data Structures; Digital Logic Design + Lab; Principles of EE I & II + Labs; Discrete Structures; Differential Equations

**In progress** Computer Architecture & Assembly + Lab; Linear Systems & Signals; Electronic Devices + Lab

## RESEARCH

**Kuiper TTS, Rutgers ECE** | Undergraduate Researcher and Team Lead (PI: Prof. Dov Kruger) Sep 2025-Present

*Python* | *PyTorch Lightning* | *Piper/VITS* | *ONNX*

- Led and helped recruit an 18-member team building offline speech for blind and low-vision readers; assigned codebase tasks
- Designed spoken-math narration rules from A/B renders of 46 real equations (“ $d q$  over  $d t$ ” became “the derivative of  $q$  with respect to  $t$ ”); adopted for spoken equations in a co-authored paper that won 2nd Place Best Full Paper, IEEE ISEC 2026
- Recorded 886 voice clips and trained custom voices on a one-command GPU pipeline; traced pitch jumps to augmentation
- Wrote audio-cleanup, augmentation and phoneme-coverage tools the team ported to C++, and the guide it uses to train voices

## EXPERIENCE

**this.is** | Lead Software Developer (Internship) Jul 2025-Aug 2026

*React* | *TypeScript* | *Firebase* | *Cloud Functions* | *Google Places API*

- Built the production web app (saved lists, search, maps, messaging) as primary engineer; mentored and managed 2 interns
- Handled a Google Places API cost spike (\$160 in 4 days, one test user) with kill switches, field masks and a daily photo budget
- Designed and built v2, a private planner for groups of 2-6 that suggests up to 3 places with stated reasons (live preview)

**Schieber Research** | Software Engineering Intern Sep 2023-Jun 2024

- Designed and built the first three prototypes of this.is, a discovery app centered on trusted friends' opinions (React, Appwrite)

**Starbucks** | Barista (part-time, alongside full-time coursework) Jul 2023-Jun 2025

## PROJECTS

**Room of Days** | Creator and Solo Developer, life-RPG habit app Jun 2026-Present

*Flutter* | *Dart* | *Swift (WidgetKit)* | *Firebase* | *TypeScript Cloud Functions* | *Codemagic CI*

- Launched on the App Store (Aug 2026) as sole developer, from design through App Review; 10+ active users, 5.0 stars (5 ratings)
- Built the XP engine: 6 life stats, anti-grind repeat decay, opt-in difficulty ladders and streak freezes, with no loss for missed days
- Deployed Firebase room sharing and an opt-in public directory, with moderation Cloud Functions and App Attest device checks

**Room Notes** | Creator and Solo Developer, cross-device notebook (private beta) Aug 2026-Present

*Flutter* | *Dart* | *Objective-C++* | *Drift/SQLite* | *Firestore* | *Cloud Functions*

- Built a notebook where iPad (Apple Pencil), phone (board capture) and laptop (equations, code) edit the same fixed-layout pages, synced through a transactional SQLite outbox; in personal use for lecture notes
- Initiated a lecturer mode on 26 Cloud Functions: QR-joined private copies, anonymous answers; deployed as a professor pilot
- Patched Flutter's iOS engine to deliver coalesced (between-frame) Apple Pencil samples; built it in CI and shipped it to TestFlight

**Knowsy Breadboard Lab, Rutgers ECE** | Team Lead (1 of 7 module leads), 3-person team May 2026-Present

*JavaScript* | *SVG* | *Verilog*

- Created a browser breadboard, wired like a real one, for Digital Logic and EE II students to prepare for lab and debug circuits
- Built live breadboard-to-schematic rendering and bench-style debugging tools: logic probe, voltmeter and wiring warnings
- Turned course materials into 8 guided labs with 90 self-checking steps; wrote a nodal-analysis circuit solver and Verilog export

**Bio (Discord bot)** | Creator and Developer Sep 2025-Present

*Node.js* | *SQLite* | *Railway*

- Created and deployed a consent-first Discord bot that helps Rutgers students find friends by shared interests; live in 2 servers

## TECHNICAL SKILLS

**Languages** Python, Dart, TypeScript, JavaScript, C/C++, Java, SQL, Verilog, Swift

**AI/ML** PyTorch Lightning, ONNX, Piper/VITS voice training, speech data preparation (VAD, loudness normalization)

**Software** Flutter, React, Node.js, Firebase (Firestore, Cloud Functions, App Check), Drift/SQLite, Git, CI/CD (Codemagic)

**Hardware and EE tools** Digital logic (Vivado), LTspice, MATLAB/Simulink, breadboarding, oscilloscopes, logic probes, soldering, Onshape CAD, 3D printing